Md. Jubayer Hossain

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PUBLICATIONS

Underline indicates mentee; *corresponding

A. Journal Articles

1. Islam, M. W., Rahman, M. M., Naznin, H., Hossain, M. S., Akter, T., Shatabde, Z. A., & ***Hossain, M. J.** (2026) "Integrative Bioinformatics Analysis Reveals COL13A1 and COL23A1 as Potential Diagnostic and Prognostic Biomarkers in Thyroid Cancer" *Health Science Reports*. http://dx.doi.org/10.1002/hsr2.72617

2. ***Hossain, M. J.**, Shahariar, M., Barsha, LHJ. Shahjahan, M., Towhid, ST., Sheikh, MK., Hasan, MM., Rahman, MH., Sazid, MS. (2026) "Lack of knowledge and training about antibiotic resistance among community pharmacists in Bangladesh: a cross-sectional study" *Health Science Reports*. http://dx.doi.org/10.1002/hsr2.72866

3. ***Hossain, M. J.**, Afrin Mim, N., Akter, N., Akter Sony, S., Haque Saikat, E., Al-Amin Hossain, M., Parvin, N., Evan, M. R., Jannat Juma, J., Parmita Roy, P., Islam Maruf, M. F., Islam, R., Labib, A., Akter Shatabde, Z., Das, M., & Amanulla, M. (2026) "Epidemiological Trends, Public Health Challenges, and Strategic Control Priorities of Dengue in Bangladesh (2000—2024): A Narrative Review" *Health Science Reports*. http://dx.doi.org/10.1002/hsr2.72747

4. Fariha, F. T. J., Fuad, M., Saha, C. S., Hossen, S., & ***Hossain, M. J.** (2025) "Comprehensive bioinformatics analysis reveals prognostic significance and immunological roles of WNT gene family in breast cancer" *Scientific Reports*. 15, 34490. 10.1038/s41598-025-13315-6

5. Ahmed, M. Z., Billah, M. M., Ferdous, J., & ***Hossain, M. J.** (2025) "Pan-cancer analysis reveals immunological and prognostic significance of CCT5 in human tumors" *Scientific Reports*. 15, 14405. 10.1038/s41598-025-88339-z

6. Bari, S. M., Fuad, M., **Hossain, M. J.**, Ahammad, I., Begum, M., & Helal, M. M. U. (2025) "A meta-analysis of public RNA-Seq data identifies conserved stress responses in rainbow trout" *BMC Genomics*. 26, 999. 10.1186/s12864-025-12127-2

7. ***Hossain, M. J.**, Das, M., Shahjahan, M., Islam, M. W., & Towhid, S. T. (2025) "Clinical and hematological manifestation of dengue patients in 2022 outbreak" *Health Science Reports*. 8, e70356. 10.1002/hsr2.70356

8. Das, M., & **Hossain, M. J.** (2025) "Young stroke in Bangladesh: Addressing rare cases with diagnostic challenges" *Stroke and Vascular Neurology*. svn-2025-004178. 10.1136/svn-2025-004178

9. ***Hossain, M. J.**, Sony, S. A., Fariha, F. T. J., & Hossen, S. (2025) "Preventing the silent threat: A perspective on preparing Bangladesh for Human Metapneumovirus" *Health Science Reports*. 8, e71101. 10.1002/hsr2.71101

10. Shanta, A. S., Islam, N., Al Asad, M., Akter, K., Habib, M. B., **Hossain, M. J.**, Nahar, S., Godman, B., & Islam, S. (2024) "Resistance and co-resistance of metallo-beta-lactamase genes in diarrheal and urinary-tract pathogens" *Microorganisms*. 12(8), 1589. 10.3390/microorganisms12081589

11. ***Hossain, M. J.**, Das, M., & Munni, U. R. (2024) "Urgent call for compulsory premarital screening for thalassemia prevention in Bangladesh" *Orphanet Journal of Rare Diseases*. 19, 326. 10.1186/s13023-024-03344-1

12. Islam, M. W., Shahjahan, M., Azad, A. K., & ***Hossain, M. J.** (2024) "Factors contributing to antibiotic misuse among parents in Dhaka City" *Scientific Reports*. 14, 2318. 10.1038/s41598-024-52313-y

13. ***Hossain, M. J.**, Das, M., Islam, M. W., Shahjahan, M., & Ferdous, J. (2024) "Community engagement and social participation in dengue prevention" *Health Science Reports*. 7, e2022. 10.1002/hsr2.2022

14. ***Hossain, M. J.**, Azad, A. K., Shahid, M. S. B., Shahjahan, M., & Ferdous, J. (2024) "Prevalence, antibiotic resistance

pattern for bacteriuria from patients with UTIs" *Health Science Reports*. 7, e2039. 10.1002/hsr2.2039

15. Akter, M. M., & *__Hossain, M. J.__ (2024) "Food consumption patterns and sedentary behaviors among university students" *Health Science Reports*. 7, e70259. 10.1002/hsr2.70259

16. *__Hossain, M. J.__, Islam, M. W., Munni, U. R., Gulshan, R., Mukta, S. A., Miah, M. S., Sultana, S., Karmakar, M., Ferdous, J., & Islam, M. A. (2023) "Health-related quality of life among thalassemia patients in Bangladesh (SF-36)" *Scientific Reports*. 13(1). 10.1038/s41598-023-34205-9

17. Towhid, S.T., __Hossain, M. J.__, Sammo, M.A.S., & Akter, S. (2022) "Perception of Students on Antibiotic Resistance and Prevention" *Eur. J. Bio. Biotech*. 3(3). 10.24018/ejbio.2022.3.3.341

18. __Hossain, M. J.__, Towhid, S.T., Sultana, S., Mukta, S.A., Gulshan, R., Miah, M.S. (2022) "Knowledge and Attitudes towards Thalassemia among Public University Students" *Microbial Bioactives*. 5(2). 10.25163/microbbioacts.526325

B. GBD Contributions

1. GBD 2023 Breast Cancer Collaborators (2026) "Global, regional, and national burden of breast cancer among females, 1990–2023, with forecasts to 2050: a systematic analysis for the Global Burden of Disease Study 2023" *The Lancet Oncology*. https://doi.org/10.1016/S1470-2045(25)00730-2

2. GBD 2023 Mental Disorder Collaborators (2026) "Updated trends in the global prevalence and burden of mental disorders, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023" *The Lancet*. https://doi.org/10.1016/S0140-6736(26)00519-2

3. GBD 2023 TB HIV Collaborators (2026) "Global, regional, and national burden of tuberculosis and multidrug-resistant tuberculosis by HIV status, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023" *The Lancet Infectious Diseases*. https://doi.org/10.1016/S1473-3099(26)00295-1

4. GBD 2023 Vaccine Coverage Collaborators (2025) "Global, regional, and national trends in routine childhood vaccination coverage from 1980 to 2023 with forecasts to 2030: A systematic analysis for the Global Burden of Disease Study 2023" *The Lancet*. https://doi.org/10.1016/S0140-6736(25)01037-2

5. GBD 2023 Headache Collaborators (2025) "Global, regional, and national burden of headache disorders, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023" *The Lancet Neurology*. https://doi.org/10.1016/S1474-4422(25)00402-8

6. GBD 2021 Musculoskeletal Disorders and COVID-19 Collaborators (2025) "Global, regional, and national burden and trend of adult musculoskeletal disorders and discrepancy between pre-pandemic and pandemic of COVID-19" *BMC Medicine (Under Review)*.

7. GBD 2023 Type 1 Diabetes Collaborators (2025) "The global burden of type 1 diabetes: a systematic analysis for the Global Burden of Disease Study 2023." *Under Review*.

8. GBD 2023 Cervical Cancer Collaborators (2025) "Global, regional and national burden of cervical cancer and associated risk factors from 1990 to 2023 and forecasts to 2050: A systematic analysis for the Global Burden of Disease Study 202" *Under Review*.

C. Papers Under Review

1. Kundu, P., Shahariar, M., Munny, F. A., Riya, A. T., Lota, H. B., Ahmed, M. Z., Meme, M. A., Kona, U. M. P., Mahmuda, F., Chowdhury, S. S., Sraboni, S. A., & *__Hossain, M. J.__ (2026) "Computational identification of Taxus baccata-derived phytochemicals targeting EZH2 to overcome therapeutic resistance in melanoma" *Scientific Reports*. https://doi.org/10.21203/rs.3.rs-8494315/v1

2. Islam, M. W., Fariha, F. T. J., Ahmed, M. Z., Ferdous, J., Lota, H. B., Hossain, M. S., Kundu, P., Shahjahan, M., & *__Hossain, M. J.__ (2025) "Bioinformatics-driven multi-omics profiling of CDK1 and CDK6 identifies prognostic and therapeutic roles in breast cancer" *Health Science Reports*.

3. Billah, M. M., Mabsurah, K., Ahammad, K., Yeana, I. J., Sumaiya, M., Islam, T., Bhattacharjee, A., Ferdous, J., &

*__Hossain, M. J.__ (2025) "Multi-omics pan-cancer analysis reveals an immunological role and prognostic potential of WDR76" *BMC Genomics*.

4. Sony, S. A., Kundu, L. R., Limon, M. H., Chowdhury, T. B. K., & & *__Hossain, M. J.__ (2025) "Predicting early antenatal care initiation at the first trimester among reproductive women in Bangladesh using machine learning" *BMC Pregnancy and Childbirth*.

D. Manuscript in Preparation

PRESENTATIONS

A. Invited Speaker

1. "Computational Biology and Bioinformatics Research in Resource-Limited Settings: Strategies, Tools and Opportunities". Jagannath University Higher Study and Research Society. Dhaka, Bangladesh. Jun 20, 2025.
2. "Mastering Biomedical Data Management". IFMSA Bangladesh. Dhaka, Bangladesh. Apr 18, 2024.
3. "Undergraduate Research - Importance, Benefits, and Challenges". CHIRAL Bangladesh. Dhaka, Bangladesh. Mar 21, 2019.
4. "State the Art of Microbial Genome Analysis". Jagannath University. Dhaka, Bangladesh. May 21, 2018.

B. Conference Presentations

1. "EBV-antigen-specific T cell clonotype fate in nasopharyngeal carcinoma: A single-cell TCR analysis", with *Rahman, M. A., Shahjahan, M., & *__Hossain, M. J.__. Dhaka University Research Society (DURS). University of Dhaka, Bangladesh. Dec 08, 2026.
2. "Reassessing the chemotherapy–TIGIT association in pancreatic cancer through cross-cohort validation", with *Chowdhury, D., Shahjahan, M., & *__Hossain, M. J.__. Dhaka University Research Society (DURS). University of Dhaka, Bangladesh. Dec 08, 2026.
3. "Chemotherapy-induced tumor cell state dynamics in osteosarcoma: NMF-based decomposition and independent bulk validation", with *Hoque, M., Shahjahan, M., & *__Hossain, M. J.__. Dhaka University Research Society (DURS). University of Dhaka, Bangladesh. Dec 08, 2026.
4. "Spatial mapping of diffuse midline glioma cell states reveals conserved territorial architecture and a mesenchymal-like interface for tumor–microenvironment crosstalk", with *Arony, T. A., Shahjahan, M., & *__Hossain, M. J.__. Dhaka University Research Society (DURS). University of Dhaka, Bangladesh. Dec 08, 2026.
5. "Cross-Cohort Meta-Analysis of Acral Melanoma Immune Cell Quantity vs. Function With Three-Way Decomposition", with *Nawar, T., Shahjahan, M., *__Hossain, M. J.__. Dhaka University Research Society (DURS). University of Dhaka, Bangladesh. Dec 08, 2026.
6. "Single-cell transcriptomic and network pharmacology insights into the therapeutic potential of Withania somnifera-derived phytochemicals against cisplatin-resistant ovarian cancer", with Islam Abhi, S. N., Meme, M. A., Shahariar, M., Barsha, L., *__Hossain, M. J.__. BEICTCON2026, IEEE. IBUAT, Dhaka, Bangladesh. Aug 20, 2026.
7. "Fezf2-Mediated Cortical Development: Multi-Omics Single-Cell Analysis", with Islam, MM, *__Hossain, M. J.__. Network of Young Biotechnologists Bangladesh. Dhaka, Bangladesh. Apr 20, 2026.
8. "Diagnostic and Electrophysiological Features of Hirayama Disease in Young Adult Male: A Case Report", with **Hossain, M. J.**, Manisha Das. Institute of Epidemiology, Disease Control and Research (IEDCR). Dhaka, Bangladesh. May 21, 2020.
9. "Quantitative Microbial Risk Assessment from Vancomycin Resistant Enterococcus faecalis and Enterococcus faecium from a Specific Neighborhood in Dhaka City, Bangladesh", with Nayeem, M.U., Mrittika, M.A., Azad, A.K., Ferdous, J., Ahmed, S., Sanyal, S.K., Towhid, S.T.. Bangladesh Society of Microbiologists. Dhaka, Bangladesh. May 21, 2020.

C. Posters

1. "Identification of Subtype-Specific Prognostic Biomarkers in Esophageal Carcinoma through Integrated Bioinformatics Analysis", with *Ahmad, F., Fuad, M., Shahjahan, M., & *__Hossain, M. J.__. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
2. "Network-Driven Bioinformatics Approach Uncovers Prognostic Signatures and Therapeutic Targets in Pancreatic Ductal Adenocarcinoma", with *Aditee, L. M., Fuad, M., Shahjahan, M., & *__Hossain, M. J.__. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
3. "Bioinformatic Analysis of Transcriptomic Data Reveals Molecular Signatures Driving Colorectal Cancer", with *Ahamed, M. S., Fuad, M., Shahjahan, M., & *__Hossain, M. J.__. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
4. "Comparative Transcriptomic Meta-Analysis Reveals Shared and Distinct Transcriptional Networks and Pathway

Crosstalk in Hepatocellular Carcinoma for Biomarker and Therapeutic Target Identification", with *Tabassum, R., Fuad, M., Shahjahan, M., & ***Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.

MEDIA COVERAGE

1. May 27, 2024, **The Business Standard**: Quoted in article What the rise in self-medication tells us about the country's healthcare system.
2. Feb 08, 2024, **The Business Standard**: Study featured in article "Parental lack of antibiotic knowledge imperils child health in Bangladesh: Study".